

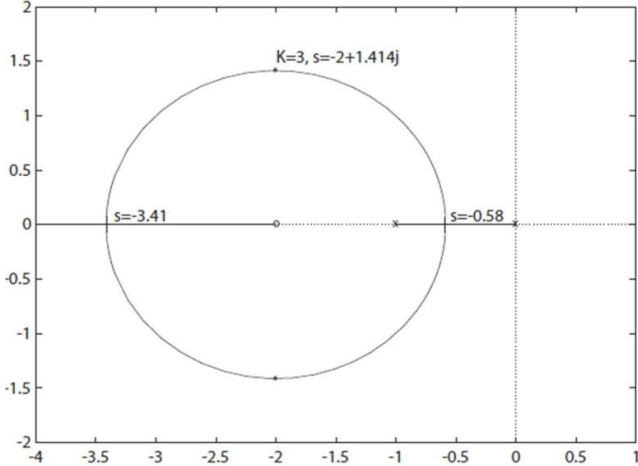
SAHRDAYA COLLEGE OF ENGINEERING & TECHNOLOGY,KODAKARA,THRISSUR-680684																						
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SIXTH SEMESTER B.TECH DEGREE(R,S)EXAMINATION, JUNE2023 (2019 Scheme)																						
Course Code: EET302																						
Course Name: LINEAR CONTROL SYSTEMS																						
Max. Marks: 100			Duration: 3 Hours																			
PART A																						
		Answer all questions, each carries 3 marks.		Marks																		
1		Definition A control system in which there is no feedback path is provided is called an <i>open loop control system</i>. Also called Open loop control system is also called non-feedback control system. Control action In open loop control system, the control action is independent of the output of the overall system. Design complexity The design and construction of an open loop control system is quite simple.	The control system in which there is a feedback path present is called a <i>closed loop control system</i>. Closed loop control system is also called a feedback control system. In closed loop control system, the control action is dependent on the output of the system. Closed loop control system has comparatively complex design and construction.	(3)																		
2		<table><thead><tr><th>LAG Network</th><th>LEAD Network</th></tr></thead><tbody><tr><td>• Improve Steady-state response</td><td>• Improve Transient response</td></tr><tr><td>• Has low pass filter characteristic</td><td>• Has high pass filter characteristic</td></tr><tr><td>• Offers unity gain to the low frequency signals and a gain of $1/\beta \ll 1$ to high frequency signal</td><td>• Offers unity gain to the low frequency signals and a gain of $1/\alpha \gg 1$ to high frequency signal</td></tr><tr><td>• Signal to noise ratio is better</td><td>• Signal to noise ratio is poorer</td></tr><tr><td>• The gain cross over frequency and the band width of the system is reduced</td><td>• The gain cross over frequency and the band width of the system is increase.</td></tr><tr><td>• Decreased bandwidth slows the response of the system</td><td>• Increased band width makes the response faster.</td></tr><tr><td>• It decreases the phase shift.</td><td>• Increases phase shift.</td></tr><tr><td>• Here pole is closer to origin than the zero.</td><td>• Here zero is closer to origin than the pole.</td></tr></tbody></table>	LAG Network	LEAD Network	• Improve Steady-state response	• Improve Transient response	• Has low pass filter characteristic	• Has high pass filter characteristic	• Offers unity gain to the low frequency signals and a gain of $1/\beta \ll 1$ to high frequency signal	• Offers unity gain to the low frequency signals and a gain of $1/\alpha \gg 1$ to high frequency signal	• Signal to noise ratio is better	• Signal to noise ratio is poorer	• The gain cross over frequency and the band width of the system is reduced	• The gain cross over frequency and the band width of the system is increase.	• Decreased bandwidth slows the response of the system	• Increased band width makes the response faster.	• It decreases the phase shift.	• Increases phase shift.	• Here pole is closer to origin than the zero.	• Here zero is closer to origin than the pole.		(3)
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Any three comparison – 3 marks																						

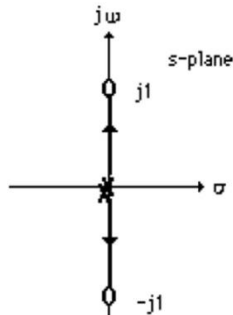
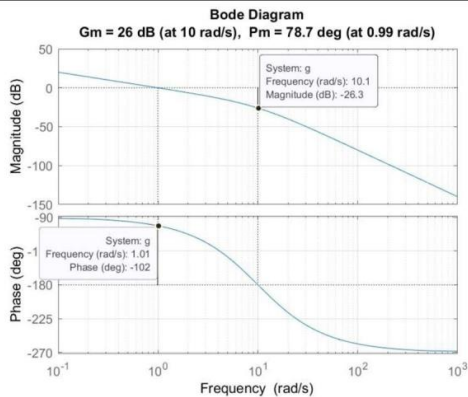
3		Three-time domain specifications for first order systems Time constant Rise time Settling time Name of any one specification and its definition – 3 marks	(3)
4		(1) Under damped system – Complex poles – 1 mark (2) Over damped system – Real poles – 1 mark (3) Critically damped system - Multiple real poles – 1 mark	(3)
5		(1) Apply Routh-Hurwitz to the closed-loop transfer function's denominator. – 1.5marks (2) Search along the imaginary axis for 180 degrees. – 1.5 marks	(3)
6		Magnitude Condition is $ G(s)H(s) = 1$ – 1.5 marks.	(3)

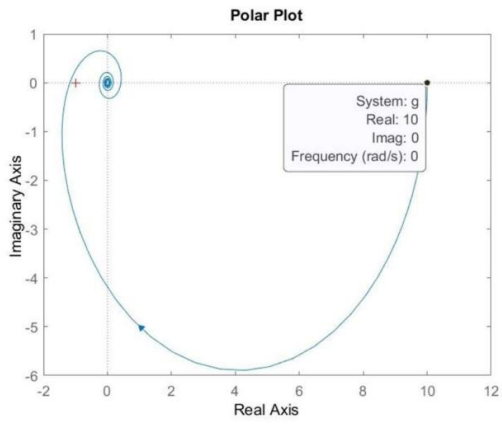
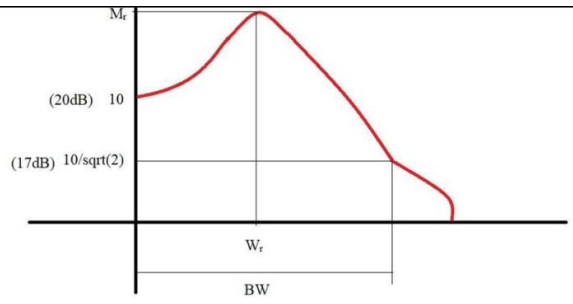
		Angle Condition is $\angle G(s)H(s) = 0^\circ \pm k360^\circ \quad (k = 0, 1, 2, \dots)$ For the positive-feedback case, the total sum of all angles from the open loop poles and zeros must be equal to $0^\circ \pm k360^\circ$. - $\angle G(s)H(s) = 0^\circ$ or multiples of 360° - 1.5 marks	
7		If the transfer function of the system is $G(s)$, let $s=j\omega$. The resulting complex number's magnitude is the magnitude response, while the resulting complex number's angle is the phase response. – 3 marks	(3)
8		Gain margin - The amount of additional open-loop gain, expressed in dB and measured at 180° of phase shift, required to make a closed-loop system unstable – 1.5 marks Phase margin - The phase margin is the amount of additional open-loop phase shift, Φ_M, required at unity gain to make the closed-loop system unstable – 1.5 marks	(3)
9		$Z = P - N$, where $Z = \#$ of closed-loop poles in the right-half plane, $P = \#$ of open-loop poles in the right-half plane, and $N = \#$ of counter-clockwise encirclements of -1 made by the mapping. – 1.5 marks Nyquist criterion tells us how many right half poles the system has and from that Whether a system is stable or not – 1.5 marks	(3)
10		A Nichols Chart is a plot used in control system design to determine the stability and closed-loop frequency response of a feedback system. Constant magnitude loci that are M-circles and constant phase angle loci that are N-circles are the fundamental components in designing the Nichols chart. The critical point in $G(j\omega)$, plane corresponds to the point of 0dB and -180° in the gain phase plane. The plot of M and N circles in the gain phase plane is known as the Nichols chart. – 3 marks	(3)
PART B			
<i>Answer one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	Circuit diagram (1 Mark) Derivation of transfer function (4 Marks) Representation of poles and zeros (1 Marks) characteristics of lead compensators- improve the transient response (1Mark)	(7)

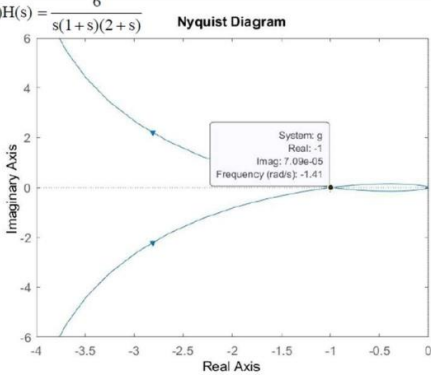
	b)	- 4 marks Brief explanation of block diagram – 3 marks (4 marks can be awarded if students either provide an explanation of the closed-loop system or draw an approximate block diagram of the general closed loop system.)	(7)
		OR	
12	a)	The transfer function for the armature-controlled DC motor Figure (1 Mark) Expression for torque, armature voltage and mechanical equation (2 Marks) Derivation of transfer function (2 Marks) The block diagram representation of the system. (1 Marks)	(6)
	b)	(i) Control Application of Tachogenerators 1. Speed control: Tachogenerators are used to measure the speed of rotating machinery, such as electric motors and turbines, and provide feedback to control systems to adjust the speed as needed. 2. Position control: Tachogenerators can also be used to measure the position of rotating machinery, such as in printing presses or assembly lines, and provide feedback for control systems to adjust the position as needed. 3. Motion control: Tachogenerators are used in systems that require precise motion control, such as in robotics and CNC machines. 4. Feedback control: Tachogenerators are used as part of a feedback control system in a wide range of applications, such as in speed controllers, servomechanisms, and automatic process control systems. - 4 marks (ii) Control application of Synchro 1. Position control: Synchros are used in systems where precise positioning control is needed, such as in the control of radar antennas and telescopes. 2. Automatic tracking: Synchros are used in systems that require automatic tracking, such as in missile guidance systems and satellite tracking systems.	(8)

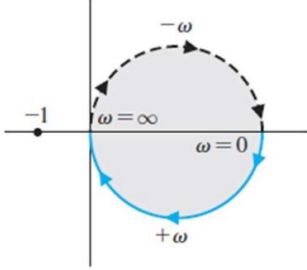
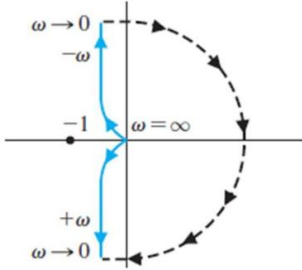
		3. Synchronization: Synchros are used to synchronize the speed and phase of multiple AC generators in a power grid, which is important for maintaining a stable power supply. 4. Control of electric motors: Synchros can be used to control the speed and direction of electric motors in industrial applications, such as in conveyor belts and cranes. - 4 marks (6 marks can be awarded for explaining the operation of Tachogenerators and Synchro)	
		Module II	
13	a)	The closed-loop transfer function is $\frac{K}{s^2 + \sqrt{2K}s + K}$. – 1 mark The damping ratio is $\zeta = \sqrt{2}/2 = 0.707$ – 1 mark and the natural frequency is $\omega_n = \sqrt{K}$. – 1 mark Therefore, (i) the percent overshoot to be $\%M_p = 4.3\%$ - 2 mark. The settling time, $T_s = 4/\zeta\omega_n = 8/\sqrt{2K}$ – 2 mark. (ii) The settling time is less than 1 second whenever $K > 32$. – 1 mark	(8)
	b)	The system is a type 0. The error constants are $K_p = 0.4$ and $K_v = 0$ – 2 mark The steady-state error to a ramp input is ∞ - 2 mark The steady-state error to a step input is $e_{ss} = 1/(1 + K_p) = 0.71$. – 2 mark	(6)
		OR	
14	a)	$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2 - \zeta^2\omega_n^2}$ $= \frac{1}{s} - \frac{(s + \zeta\omega_n) + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2} = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta\omega_n}{\omega_n\sqrt{1 - \zeta^2}} \omega_n\sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2}$ Hence, $c(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n\sqrt{1 - \zeta^2} t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_n\sqrt{1 - \zeta^2} t \right)$ $= 1 - e^{-\zeta\omega_n t} \sqrt{1 + \frac{\zeta^2}{1 - \zeta^2}} \cos (\omega_n\sqrt{1 - \zeta^2} t - \phi) = 1 - e^{-\zeta\omega_n t} \frac{1}{\sqrt{1 - \zeta^2}} \cos (\omega_n\sqrt{1 - \zeta^2} t - \phi),$ where $\phi = \tan^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}}$ - 7 marks - (Marks may be given for steps) - (50% marks for partial fraction expansion)	(7)

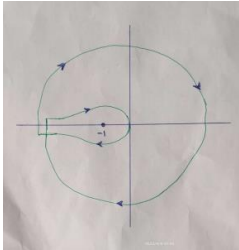
	b)	the Routh array $ \begin{array}{c ccc} s^5 & 1 & 2 & 1 \\ s^4 & 1 & 1 & K \\ s^3 & 1 & 1-K & \\ s^2 & K & K & \\ s^1 & -K & 0 & \\ s^0 & K & & \end{array} $ - 4 marks Examining the first column, we determine that for stability we need $K > 0$ and $K < 0$. Therefore, the system is unstable for all K. - 3 marks	(7)
		Module III	
15	a)	Centroid = -1 - 1 mark Angle of Asymptotes = -180° - 1 mark Break away/in points = -0.58 & -3/.41 - 2 marks Drawing the root locus - 5 Marks The gain when the closed loop system roots are located at $s_{1,2} = -2 \pm \sqrt{2}j$ is $K = 3$ - 1 mark 	(10)
	b)	Effect of adding poles 1. Root locus shift towards imaginary axis 2. Relative stability decreases 3. Settling time increases - 2 Marks Effect adding zeros.	(4)

		1. Root locus shift towards left of s plane 2. Relative stability increases 3. Settling time decreases - 2 Marks	
		OR	
16	a)	No. of poles = No. of zeros – No Centroid and angle of Asymptotes. Break Away Point = 0 No imaginary axis crossing point. – 3 marks Drawing the root locus - 5 Marks  The closed loop system is marginally stable for all values of K – 2 marks	(10)
	b)	The general steps involved in the design of lag compensator using root locus- Explain the step-by-step methods for design (4 Marks)	(4)
		Module IV	
17	a)	 Drawing gain plot - 3 Marks Drawing phase plot - 3 Marks With asymptotic Bode plots, Gain cross over frequency = 1r/s Phase Margin = 78 degree (1 mark)	(10)

		Phase cross over frequency = 10r/s Gain Margin = 20 dB (1 mark) Both Gain Margin and Phase Margin are positive. Hence system is stable. (2 marks)	
	b)	For time delay. $G(j\omega) = e^{-j\omega T}$ Magnitude of $G(j\omega) = 1$ hence No change in magnitude plot – 2 marks Phase of $G(j\omega) = -\omega T$ (radians) = $-57.3\omega T$ (degree), hence System has an excessive phase lag at high frequencies – 2 marks	(4)
		OR	
18	a)	$G(s)H(s) = \frac{10e^{-0.2s}}{(s+1)}$ $G(j\omega)H(j\omega) = \frac{10e^{-0.2j\omega}}{(j\omega+1)}$ $= \frac{10}{\sqrt{1+\omega^2}} \angle -(0.2 * 57.3\omega + \tan^{-1} \omega)$ - 4 marks  Polar Plot Drawing the polar plot - 6 marks	(10)
	b)		(4)

		Resonant Frequency (ω_r): It is the frequency at which the maximum magnitude occurs. Resonant peak (M_r): It is the maximum magnitude of closed loop system. Band Width (BW): It is the range of frequencies over which system acts satisfactory. Cut off rate: It is the rate at which the magnitude falls at high frequencies. - 4 marks	
		Module V	
19	a)	$G(s)H(s) = \frac{6}{s(1+s)(2+s)}$ Nyquist Diagram  Drawing the Nyquist plot - 6 Marks Identifying the crossing point exactly at (-1, 0) with $N=0$, $P=0$ $\square N=P-Z$ & $Z=0$ and the system is oscillatory – 4 marks Instead of the above, if the student identifies a crossing point away from (-1, 0) with a respective conclusion, 2 marks may be awarded alongwith the marks for sketching the Nyquist plot.	(10)
	b)	1. Phase Margin: The phase margin is a measure of the amount of phase lag in the system at the gain crossover frequency. A good rule of thumb is to design the lag compensator to provide a phase margin of at least 45 degrees to ensure stability. The corner frequency of the lag compensator should be selected to achieve this phase margin requirement. 2. Steady-state error: Lag compensators are used to reduce steady-state error in a control system. The corner frequency of the lag compensator should be selected based on the desired steady-state error reduction. A lower corner frequency will result in a greater steady-state error reduction. 3. Bandwidth: The bandwidth of the control system is the range of frequencies over which the system can effectively track changes in the reference signal. The	(4)

		corner frequency of the lag compensator should be selected to maintain the desired bandwidth of the system. A lower corner frequency will result in a lower bandwidth. 4. Sensitivity: The sensitivity of the control system is a measure of how much the system output changes in response to changes in the system parameters. The corner frequency of the lag compensator should be selected to minimize the sensitivity of the system to changes in the plant parameters. - 4 marks	
		OR	
20	a)	(i) $G(s) = \frac{K}{sT_1 + 1}$  Sketching Nyquist plot – 2 marks Stable for all values of K - 1 mark (ii) $G(s) = \frac{K}{s(sT_1 + 1)}$  Sketching Nyquist plot – 2 marks Stable for all values of K - 1 mark	(9)

	$(iii) G(s) = \frac{K}{s^2(sT_1 + 1)}$  Sketching Nyquist plot – 2 marks For all values of K, system is unstable - 1 mark	
b)	The general steps involved in the design of lead compensator using bode plot- Explain the step-by-step methods for design (5 Marks)	(5)
